

```
## Introduction to Python
# Python is a high-level, interpreted, and object-oriented programming language
# that is easy to read, write, and understand. It is widely used for software development,
# web development, data science, machine learning, automation, and scripting
```

```
# Fundamentals of Python
# (1). Data type
# (2). Variable
# (3). print() and input()
```

```
# (1). Data type
# (a). Integer ---> All positive or negative values .
# Ex. 34 , 67 , -12

# (b). Float ---> All decimal values .
# Ex. 12.34 , 67.89

# (c). String ---> ' ' , " " , ' ' ' '
# Ex. 'sam' , "sam" , ' ' 'sam' ' '

# (d). Boolean ---> True , False
```

```
# type() ----> It will return data type of element .
type(45)
type(35.78)
type('sam')
type(5>4)
```

```
bool
```

```
# Variable ---> This is a container in which we can store or assign our values .
```

```
# (3). Input() -----> It will take query from users and by default its output will be in
# string format .
```

```
a = 1
b = 2
c = a + b
print(c)
```

```
3
```

```
a = '1'
b = '2'
c = a + b
print(c)
```

```
12
```

```
a = input("Enter first number:")
b = input("Enter second number:")
c = a + b
print("Total=" , c)
```

```
Enter first number:1
Enter second number:2
Total= 12
```

```
# type-casting ----> Using this method , we can change or update data type .
```

```
a = '12'
int(a)
float(a)
```

```
12.0
```

```
a = int(input("Enter first number:"))
b = int(input("Enter second number:"))
c = a + b
print("Total=" , c)
```

```
Enter first number:1
Enter second number:2
```

```
Total= 3
```

```
a = input("Enter first number:")
b = input("Enter second number:")
c = int(a) + int(b)
print("Total=" , c)
```

```
Enter first number:1
Enter second number:2
Total= 3
```

```
# Mostly Used Operators in Python
# (1) Addition(+) ----> 2+3 = 5
# (2). Subtraction(-) ----> 5-2=3
# (3). Multiplication(*) ----> 2*5 = 10
# (4). Division(/) ----> 5.2 = 2.5
# (5). Floor Division(//) --> It will convert all the values in round figure . ----> 5//2 = 2
# (6). Module(%) ----> It is used for remainder calculation . ----> 25%10= 5(Remainder)
# (7). Equal to(=) ----> It is used for assign the values . a = 5
# (8). Equal Equal to(==) ----> It is used for comparioson of 2 values . ----> 'sam'=='raj' , 4==5
# (9). Not Equal to(!=) ----> 4!=5 , 'sam'!='mohit'
# (10).Greater than(>) ----> 5>4
# (11). Less than (<) ----> 4<8
# (12). Greater than or Equal to(>=) ----> 5>=5
# (13). Less than or Equal to(<=) ----> 4<=5
```

```
# Conditional Statement ---->
# if
# elif <-----> User_query
# else
```

```
a = "sam"
if(a=="sam"):
    print("Yes")
elif(a=="raj"):
    print("No")
elif(a=="rohit"):
    print("ok")
else:
    print("Try Again...")
```

```
Yes
```

```
# Q.2 Create a program who will tell user is child , young , old .
```

```
a = int(input("Enter number:"))
if(0<=a<=20):
    print("Child")
elif(21<=a<=50):
    print("Young")
elif(51<=a<=80):
    print("Old")
else:
    print("Please enter a valid age....")
```

```
Enter number:78
Old
```

```
# Nested Conditional Statement ---->
print("Pizza:100" , "burgur:50")
a = input("Enter you item:")
if(a == "Pizza"):
    print("Your yummy item is Pizza")
    b = int(input("Enter your quantity:"))
    c = b*100
    if(c>1000):
        print("Your 10% discounted price is :" , c*0.9)
    else:
        print("Your total payment is :" , b*100)

elif(a== "burgur"):
    print("Yooour yummy item is burgur")
else:
    print("Select a valid Menu....")
```

```
Pizza:100 burgur:50
Enter you item:Pizza
Your yummy item is pizza
Enter your quantity:5
```

```
Your total payment is : 500
```

```
a = 100  
b = a*0.9  
b
```

```
90.0
```

Start coding or [generate](#) with AI.

```
# Councelling system ---> usser course(python, java , js) ---> course ---> fees , time duration .
```